Scale Invariant Topological Constraint and First-Principle Calibration of Song Unit (Sg) in ANG-TOE Theory

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Abstract

Current fundamental physical constants are empirically measured quantities lacking first-principle theoretical origin. Based on the core zero-global-angular-momentum axiom of ANG-TOE, this paper proves that the universe's primordial six-dimensional angular momentum manifold possesses **exact global scale invariance**. A single zero-sum topological constraint cannot fix the absolute dimensional scale of the universe, leading to a free scaling degree of freedom. This study derives the complete scale transformation law of angular momentum, force, mass, space, and time observables, establishes the invariant dimensionless constraint equations governing the Song fundamental unit (Sg), and clarifies the theoretical distinction between **pure axiomatic scale freedom** and **observational projective scale calibration**. This work demonstrates that the Planck constant is merely the low-dimensional projective reading of the Sg quantum under the specific topological scaling configuration of our perceptual spacetime, rather than an inherent universal constant. A complete closed first-principle scheme for solving the absolute value of Sg is proposed, providing the fundamental unit foundation for the unified self-consistent physical system of ANG-TOE.

Keywords: ANG-TOE; zero angular momentum axiom; scale invariance; topological projection; Song unit (Sg); fundamental constant origin

1. Introduction

Traditional physics treats Planck constant, speed of light, and mass scale as irreducible fundamental constants artificially embedded in spacetime background geometry. No existing theory can derive their absolute values from pure first principles.

The core difficulty lies in:

- Conventional theories presuppose dimensional scales of spacetime in advance;

- No existing axiomatic framework can explain why fundamental constants possess specific numerical magnitudes;
- All constant values rely entirely on experimental fitting with no topological origin interpretation.

The ANG-TOE system constructs the universe from a single primordial axiom: total cosmic angular momentum is permanently zero. All macroscopic physical quantities are emergent projective effects of high-dimensional topological networks. This leads to a critical theoretical conclusion:

> The zero-sum angular momentum constraint is a **topological constraint without dimensional scale**, therefore the primitive universe has no inherent fixed scale, and all dimensional constants contain global scaling degrees of freedom.

This paper systematically establishes the scale invariance theory of the ANG-TOE framework, derives the full set of invariant equations for the Song unit (Sg), and completes the theoretical positioning between pure first-principle freedom and observational calibration.

2. Primitive Axiom and Global Scale Symmetry

2.1 Core Axiom

The total angular momentum of the closed cosmic manifold satisfies:

$$\boldsymbol{J}_{\text{total}} \equiv \mathbf{0}$$

This axiom is a topological zero-balance constraint, containing no length dimension, no time dimension, and no unit scale information.

2.2 Global Scaling Transformation Definition

Define global topological scaling factor $\lambda > 0$ for the six-dimensional primordial angular momentum manifold $\mathcal{M}_6^{\text{phys}}$.

Perform uniform scaling on all intrinsic angular momentum links of the manifold:

$$\boldsymbol{J} \to \lambda \boldsymbol{J}$$

Since the zero vector satisfies scale invariance:

$$\lambda \cdot \mathbf{0} \equiv \mathbf{0}$$

Key Theorem:

> The zero angular momentum axiom holds for **any arbitrary scaling factor** λ. The primitive universe possesses exact continuous global scale invariance, and the single axiom cannot eliminate scaling degeneracy.

This is the fundamental theoretical root why all dimensional physical constants cannot be uniquely solved by Axiom 0 alone.

3. Scale Transformation Rules of Projective Observables

Under global scaling λ, all low-dimensional projective physical quantities derived from angular momentum networks follow strict linear scaling laws.

3.1 Fundamental Quantity Scaling

Under global scaling transformation, core physical quantities obey the following linear scaling laws:

$$\begin{aligned} \boldsymbol{J}_3 &\to \lambda \boldsymbol{J}_3 \\ \boldsymbol{F} &\to \lambda \boldsymbol{F} \\ m &\to \lambda m \\ dl &\to \lambda dl \\ dt &\to \lambda dt \end{aligned}$$

3.2 Invariance of Spacetime Coupling Constant (Speed of Light)

The speed of light is the ratio of spatial differential and temporal differential:

$$c = \frac{dl}{dt} \to \frac{\lambda dl}{\lambda dt} = c$$

Conclusion:

- c is a **scale-invariant geometric ratio constant**;
- The speed of light contains no scaling freedom and is uniquely determined by the intrinsic projection geometry of the manifold;
- The scaling freedom of the entire physical system **only concentrates on the angular momentum quantum scale**.

4. Scale Invariant Constraint Equations of Song Unit (Sg)

4.1 Definition of Song Fundamental Unit

Sg is defined as the **minimum topological quantum unit of angular momentum exchange** in the cosmic network, the discrete smallest eigenvalue of primordial angular momentum topology.

Dimension of Sg:

$$[\mathrm{Sg}] = \mathrm{kg} \cdot \mathrm{m}^2 \cdot \mathrm{s}^{-1}$$

4.2 Core Invariant Relation Derivation

From the projective emergence mechanism of mass, space, and angular momentum, the dimensionless topological invariant Ψ (topological closure coefficient) is introduced:

$$\Psi = \frac{mcdl}{\mathrm{Sg}}$$

Ψ is a pure topological dimensionless parameter, independent of global scaling and unchanged under λ transformation.

Rearranged to obtain the **first-principle closed expression of Sg**:

$$\mathrm{Sg} = \Psi \cdot mcdl$$

This is the most fundamental invariant equation of dimensional scale in ANG-TOE theory.

4.3 Complete Scale Invariant System

The full set of governing equations for cosmic scale freedom:

$$\begin{cases} \mathrm{Sg} \propto \lambda \\ m \propto \lambda \\ dl \propto \lambda \\ dt \propto \lambda \\ c = \text{invariant} \\ \Psi = \text{invariant} \end{cases}$$

Physical interpretation:

1. All dimensional scales of the universe originate from the same global scaling factor λ;

2. Speed of light and topological closure rate are pure geometric invariants with no scale degeneracy;

3. The entire dimensional system of physics has **only one single free degree of freedom**, namely the angular momentum quantum scale λ.

5. Theoretical Essence: Why Planck Constant Is Not Fundamental

In the standard physical system, the Planck constant h is artificially regarded as a cosmic fundamental constant.

This paper strictly proves:

5.1 Projection Calibration Principle

Under the fixed topological projection configuration of our observable universe ($\lambda = \lambda_0$):

$\mathrm{Sg}|_{\lambda=\lambda_0} = h = 6.62607015 \times 10^{-34}\mathrm{J} \cdot \mathrm{s}$

5.2 Critical New Physical Conclusion

1. h **is not a universal constant of the primitive universe**, but the **specific numerical reading of Sg under our projective scale**;

2. If the cosmic global topological scaling factor λ changes synchronously, all mass scales, length scales, and quantum scales of the universe will scale synchronously, while all physical laws and dimensionless invariants remain completely unchanged;

3. The reason why human observers measure fixed h is that **we are in a fixed projection scale frame and scale synchronously with the entire universe**, resulting in invariant observational reading.

6. Two Paradigms for Sg Value Determination

6.1 Paradigm One: Observational Calibration (Empirical Matching)

Based on existing experimental data of the current universe's projection scale, lock λ_0 and obtain the Sg numerical value.

This paradigm is consistent with all existing quantum experiments and fully compatible with quantum mechanics phenomena.

6.2 Paradigm Two: Pure First-Principle Closed Solution (Complete TOE)

To completely eliminate scaling freedom and solve the absolute theoretical value of Sg without relying on any experiment, it is necessary to add one independent topological scale axiom:

> **Axiom 1 (Minimum Topological Closed Loop Boundary)**
> The primordial angular momentum network has a minimum non-zero closed topological loop perimeter, which defines the absolute scale boundary of the universe.

After supplementing this axiom, the system of equations is fully closed, all scaling freedom disappears, and the absolute numerical value of Sg can be strictly derived from first principles without any empirical parameters.

7. Discussion: The Hierarchical Subversion of Fundamental Constant Philosophy

Traditional physics philosophy:

> Multiple independent fundamental constants (c, h, G) are preset by the universe, with no internal connection.

ANG-TOE scale invariance philosophy:

1. The universe has only **one scaling degree of freedom** in essence;
2. The speed of light is a geometric ratio invariant;
3. All dimensional constant differences originate from the synchronous scaling projection of the same topological origin;
4. All empirical fundamental constants are **derived projective quantities**, not primitive preset quantities.

This fundamentally eliminates the artificial multiplicity of cosmic fundamental constants and realizes the unification of constant origins under a single topological axiom.

8. Conclusion

1. The zero global angular momentum axiom of ANG-TOE leads to inherent continuous global scale invariance of the cosmic manifold, bringing a single universal scaling free degree of freedom;
2. This paper derives the complete scaling transformation law of angular momentum, force, mass, spacetime, and the strict invariant governing equation of the Song unit $\mathrm{Sg} = \Psi mcdl$;
3. The Planck constant is proved to be the observational reading of the Sg fundamental quantum under the current topological projection scale, not an inherent cosmic constant;
4. The theoretical system clearly distinguishes empirical calibration and pure first-principle closed solution paths, laying the core unit theoretical foundation for the complete axiomatic TOE system.

This work resolves the longstanding fundamental problem in theoretical physics: **the origin and uniqueness of fundamental physical scales**, and establishes the self-consistent unit ontology of ANG-TOE.

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